

PV–PORT 180 A port board

Rev F0 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | Ports A (PV) and B (bus/battery) of PV–P100/110, common negative BUS–

01 Connectors and sensor supply

PORT 26–way to CTRL–C2000, 14–way to SYS–IO–AUX, ID + hold readback, +24V → +3V3S buck, AGND tie, PE bond

File: sheets/01_io.kicad_sch

02 Port A power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/02_portA_power.kicad_sch

03 Port A sensing and hold logic

VA bus side, VAX terminal side (AMC3330), IA current (AMC3302), two–channel hold logic, proof–test injection, readback RB1/RB2

File: sheets/03_portA_sense.kicad_sch

04 Port B power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/04_portB_power.kicad_sch

05 Port B sensing and hold logic

VB bus side, VBX terminal side (AMC3330), IB current (AMC3302), two–channel hold logic, proof–test injection, readback RB1/RB2

File: sheets/05_portB_sense.kicad_sch

06 Insulation measurement

AUX1 = V(PE–BUS–), AUX2 = V(PE–A+) (AMC3330); 992 k test strings A+ → PE, BUS– → PE, each with a common–source SiC pair (TPSI3050)

File: sheets/06_imd.kicad_sch

Common negative: port A and port B share BUS– (non–isolated buck–boost), so both ports are one HV domain; the main contactors switch the + pole only, fuses are in both poles. Chassis–mounted parts (terminals, fuses, CM choke, contactors, RST 200 resistors, thermostats, shunt) are drawn here with their real MPN or an RFQ / CUSTOM specification.

Isolators (checked on every build): AMC3330/AMC3302, TPSI3050–Q1, CNY65B, contactor and relay coil–contact. Declared HV–PE crossings: MOV3 of each varistor Y, Y capacitors, the PE–side SiC of each insulation–measurement pair, the PE–end element of the AUX dividers.

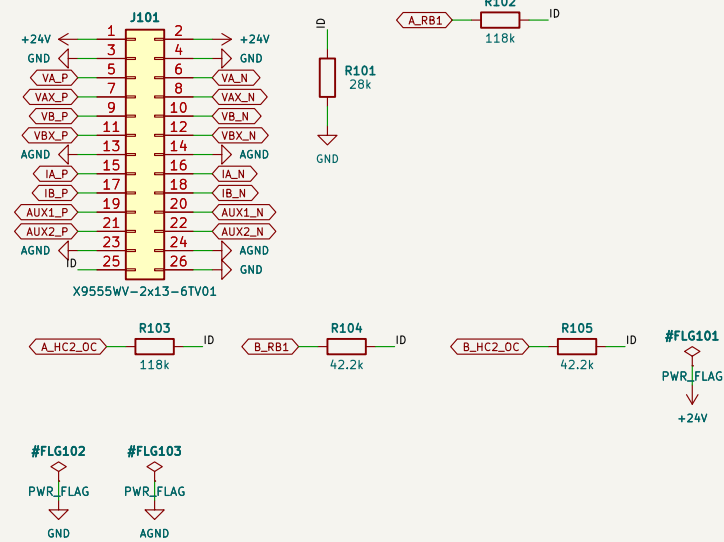
IC–17/IC–18: the whole board is conformal–coated to PD1 (IEC 60664–3 Type 1) over all HV copper and every HV–PELV isolator footprint (8–8.5 mm packages < 10 mm reinforced at PD2). Only the NH fuse links / bases and the DC terminals sit in the cooling–air path (PD3); the board, varistor network and branch fuses are in the closed PD2 compartment.

Surge credit (D–032, IC–01/02/03): the on–board varistor Y limits pole–PE and pole–pole to 3.8 kV at In 3 kA (calculated) and is monitored: reinforced is dimensioned for 6 kV (AMC3302 6.25 kV, AMC3330 7.7 kV pass on this case only). IC–04 OPEN: the G7L coil–contact rating (4000 V AC) is below reinforced – fix in rev F1 (own isolated coil supply).

SYS–IO–AUX: K_DISCH energised first at power–up; the port block itself inhibits the discharge of a closed port. Coil flyback never reaches the DO clamps in normal operation (two–switch coil drives, varistor / zener across each coil).

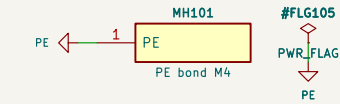
Not bench–validated. Values: sim/port_design.py → port_spec.json		
Sheet 1		
Schematics authored by Chinmoy Bhuyan		
HV–PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV–PE basic + bonding		
Sheet: /		
File: PV–PORT–180.kicad_sch		
Title: PV–PORT 180 A port board – Cover / Hierarchy		
Size: A3	Date: 2026–10–04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 1/7

PORT CONNECTOR (CTRL-C2000) + ID WITH HOLD READBACK



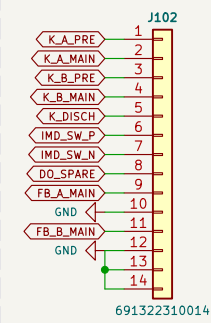
interfaces.PORT, ribbon 1:1. AMC outputs drive the cable directly (≤ 500 pF, AMC3330 p4); anti-alias RC is on CTRL-C2000. ID: 28k || readback network = 9.99 k (PV-PORT code 10.0k; CTRL-C2000 rev D values). RB outputs through 2 x 118k (port A) and 2 x 42.2k (port B): nine states, ≥ 155 counts apart (81 with the hold coils on, sim/port_design.py sec. 7.4); CTRL decodes them with its port-only table.

PE BOND



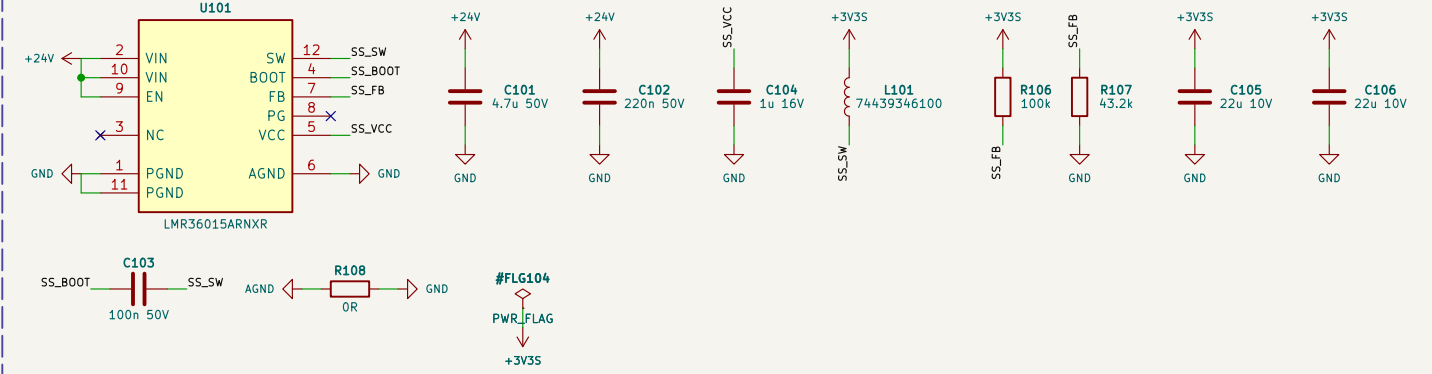
Board PE to chassis at the SPD / Y-capacitor end.

CONTROL CONNECTOR (SYS-IO-AUX)



port.CTRL_PINS: DO1-DO8 (DO_SPARE = proof test), FB_A + its GND, FB_B + its GND (INT-02), 2 coil returns. Port B (battery) drives 2 HVC43 coils: K_B_MAIN up to 0.60 A -> SYS DO4 needs ILIM 10 k (1.58-2.30 A).

SENSOR SUPPLY +3V3S



LMR36015 400 kHz, 24 V -> 3.315 V (100 k / 43.2 k, SNVSB49D sec 10.2), <= 0.34 A for
6 x AMC3330 + 2 x AMC3302 + hold logic. AGND joins GND only here (0 ohm).

Not bench-validated. Values: `sim/port_design.py -> port_spec.json`
Sheet 2
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV-PE basic + bonding

Sheet: /01 Connectors and sensor supply/
File: 01_io.kicad_sch

Title: PV-PORT 180 A port board – Connectors and sensor supply

Size: A3	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 2/7

[illegible]

The diagram illustrates the typical application circuit for the TNPV12061M00BEEA. It features a central U302 component (AMC3330WE) with 16 pins. The circuit is powered by +3V3 and +3V5. The output is VAX_D0. The circuit includes various resistors (R309, R310, R311, R312, R313, R314) and capacitors (C310, C311, C312, C313, C314, C315, C316, C317, C318) connected to the pins of the U302. The circuit is labeled 'U302' and 'AMC3330WE'.

$|I_A| > 410 \text{ A}$ (band 382–441 A, own REF3030E + 0.1 % network, CM-cancelling) AND FB = closed AND
 ROAD = valid $\rightarrow$ HOLD1. Test injection 86.6 μ from A_TST_T (side A) / A_TST_TD (side B): shifts the threshold
 by 459 A, so a healthy side trips at 0 A under TEST only.

[illegible]

100 k / 15 k -> Schmitt buffer T (ch1 side A). T -> 47 k / 22 uF -> T_d (ch1 side B). One-shots (3.3 uF / 47 k) of T -> T2A (ch2 side A) and of T_d -> T2B (ch2 side B). No single fault trips both channels for longer than one one-shot. Firmware runs it at every start, contactor open and closed (report sec. 7.4; windows, what it finds and what it does not).

[illegible]

Not bench-validated. Values: sim/port_design.py -> port_spec.json

Sheet 4

Schematics authored by Chinmoy Bhuyan

HV-PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV-PE basic + bonding

Sheet: /03 Port A sensing and hold logic/
File: 03_portA_sense.kicad_sch

Title: PV-PORT 180 A port board – Port A sensing and hold logic

Size: A2	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 4/7

The schematic diagram illustrates the AMC3330DWE chip and its external components. The chip is a yellow rectangle with pins 1-16 on the left and right. Internal blocks include a bus interface (BUS+), a DCDC converter, and a VDD/LDO regulator. External components include capacitors C501-C509 and resistors R501-R506. The chip is labeled 'AMC3330DWE' at the bottom.

Pin Connections:

- Pin 1:** VB_IN
- Pin 2:** BUS+
- Pin 3:** VB_DCO
- Pin 4:** VB_HLDO
- Pin 5:** VB_HLDO
- Pin 6:** INP
- Pin 7:** INN
- Pin 8:** DCDC_OUT
- Pin 9:** HLDO_IN
- Pin 10:** HLDO_OUT
- Pin 11:** NC
- Pin 12:** DCDC_HGND
- Pin 13:** DCDC_GND
- Pin 14:** GND
- Pin 15:** OUTP
- Pin 16:** OUTN

Internal Blocks:

- BUS+:** Connected to pins 2 and 8.
- DCDC:** Connected to pins 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16.
- VDD/LDO:** Connected to pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16.

External Components:

- Resistors:** R501, R502, R503, R504, R505, R506.
- Capacitors:** C501, C502, C503, C504, C505, C506, C507, C508, C509.

Power and Ground Connections:

- +3V35:** Connected to pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16.
- AGND:** Connected to pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16.

6 x TNPV1206 1.00 M (167 V each at 1000 V) + 4.99 k: 1 V = 1203 V, clip 1504 V. 4.7 nF: lag <= 27 us incl. AMC3330. Out: VB_P/N (+/-2 V on 1.44 V, PORT contract). HGND = BUS-.

[illegible]

6 x TNPV1206 1.00 M (167 V each at 1000 V) + 4.99 k: 1 V = 1203 V, clip 1504 V, 4.7 nF: lag
 <= 27 us incl. AMC3330. Out: VBX_P/N (+/-2 V on 1.44 V, PORT contract). HGND = BUS-.

[illegible]

Linear ± 500 A, clip ± 640 A (OC trip 270 A, SC trip 450 A inside). 10 ohm + 10 nF input filter (SBASA11B Fig 8-2). HGND on the shunt's X- bolt by its own trace. Fail-safe out = -2.57 V

|IB| > 410 A (band 382-441 A, own REF3030E + 0.1 % network, CM-cancelling) AND FB = closed AND DIAG = valid -> HOLD1. Test injection 86.6 k from B_TST_T (side A) / B_TST_TD (side B): shifts the threshold by 459 A, so a healthy side trips at 0 A under TEST only.

|B| > 410 A (band 382-441 A, own REF3030E + 0.1 % network, CM-cancelling) AND FB = closed AND
DIAG = valid -> HOLD2. Test injection 86.6 k from B_TST_T2A (side A) / B_TST_T2B (side B): shifts the threshold
by 459 A, so a healthy side trips at 0 A under TEST only.

The schematic diagram illustrates the internal circuitry of the B-TST module, featuring five op-amp comparators (U512, U513, U514, U515, U516) and their associated passive components. Each comparator is configured with VCC at pin 1, AGND at pin 5, and NC at pin 4. The outputs (pins 2 and 3) are connected to various test points (B_TST_T, B_TST_DRC, B_TST_XA, B_TST_XAI, B_TST_TD, B_TST_XB, B_TST_XBI, B_TST_XB) through resistors (R549, R551, R554, R552, R548) and capacitors (C538, C539, C540, C542, C544, C541). The module is powered by +3V3S and AGND.

100 k / 15 k -> Schmitt buffer T (ch1 side A), T -> 47 k / 22 uF -> T_d (ch1 side B). One-shot (3.3 uF / 47 k) of T -> T2A (ch2 side A) and of T_d -> T2B (ch2 side B). No single fault trips both channels for longer than one one-shot. Firmware runs it at every start, contactor open and closed (report sec. 7.4: windows, what it finds and what it does not).

[illegible]

RB1 = OC1. OR hold high-side output present (100 k / 15 k); at rest RB1 = 1 means a stuck channel 1.
RB2 = OC2. PV-PORT sums them onto the ID line (board_io); firmware reads RB1 + RB2 against the measured current in operation and through the proof-test windows at start.

Not bench-validated. Values: `sim/port_design.py -> port_spec.json`
Sheet 6
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV-PE basic + bonding

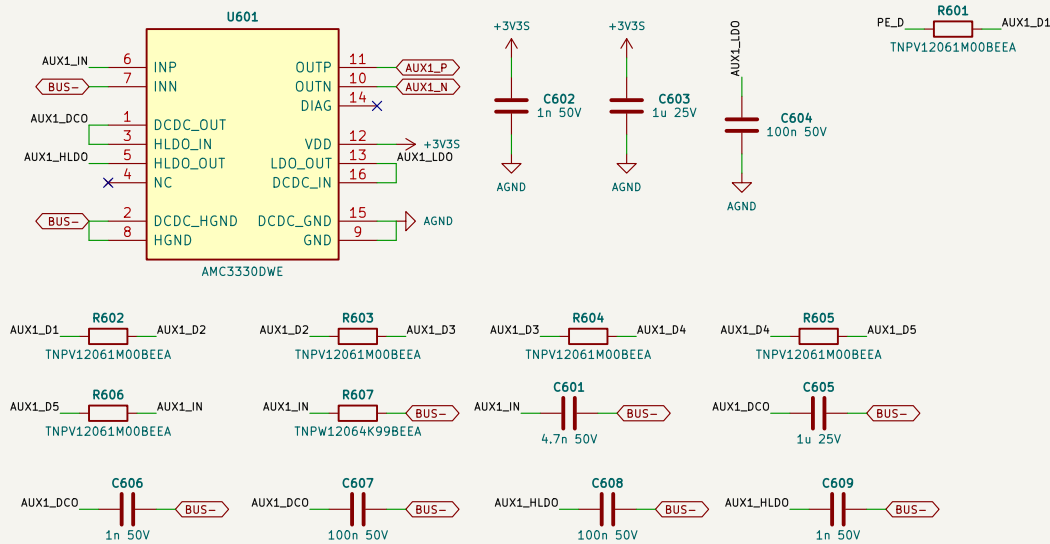
Sheet: /05 Port B sensing and hold logic/
File: 05_portB_sense.kicad_sch

Title: PV-PORT 180 A port board – Port B sensing and hold logic

Size: A2	Date: 2026-10-04
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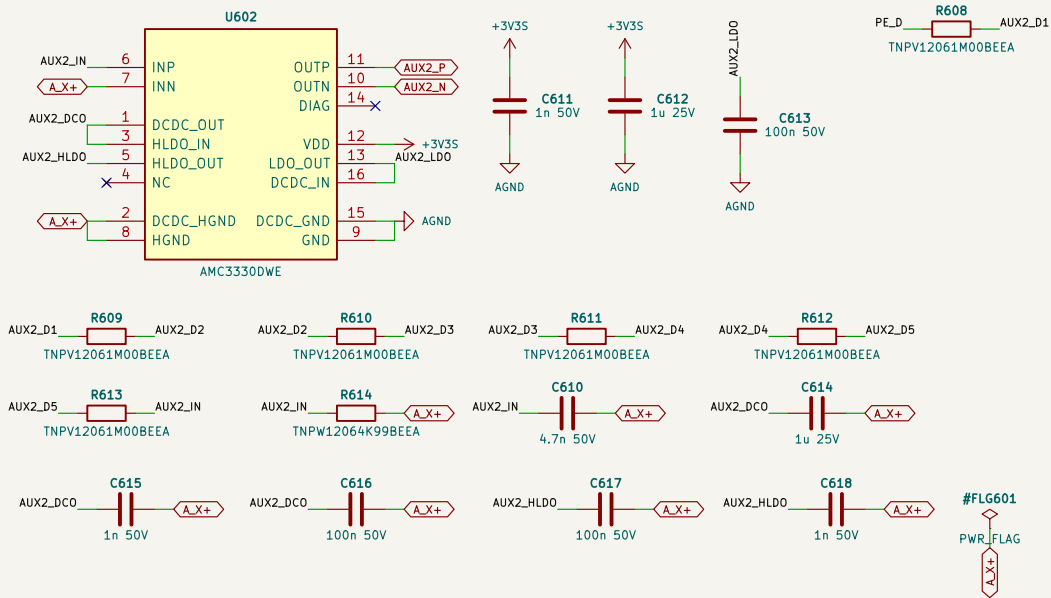
Rev: F0

AUX1 – PE RELATIVE TO BUS–



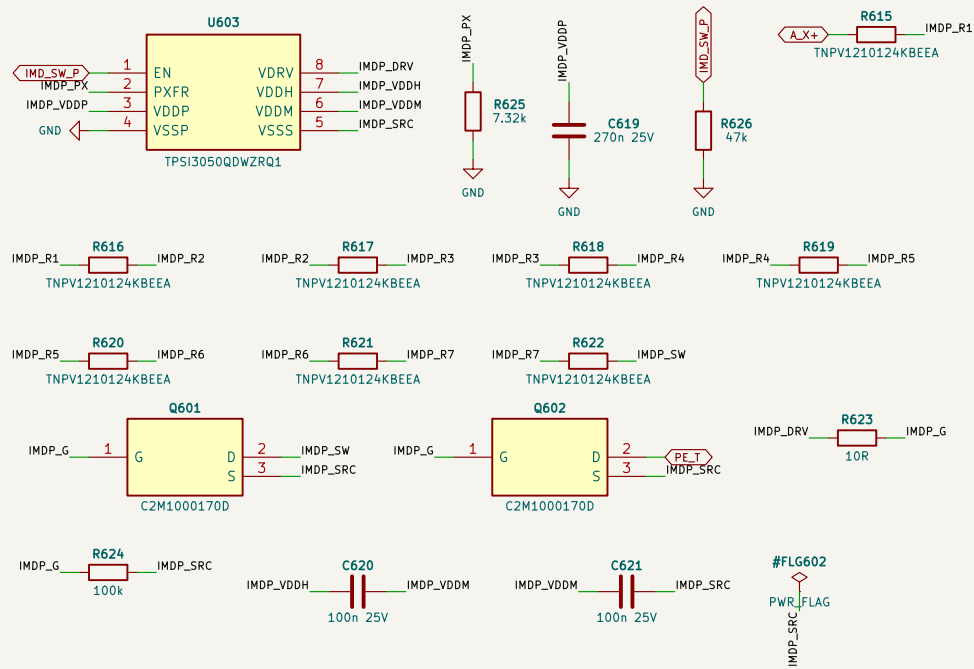
6 x 1.00 M from PE + 4.99 k: +/-1203 V linear. The divider is a fixed, known 6.005 M path to PE (included in the IMD equations, sim/port_design.py sec. 8). First element = HV–PE crossing.

AUX2 – PE RELATIVE TO A_X+



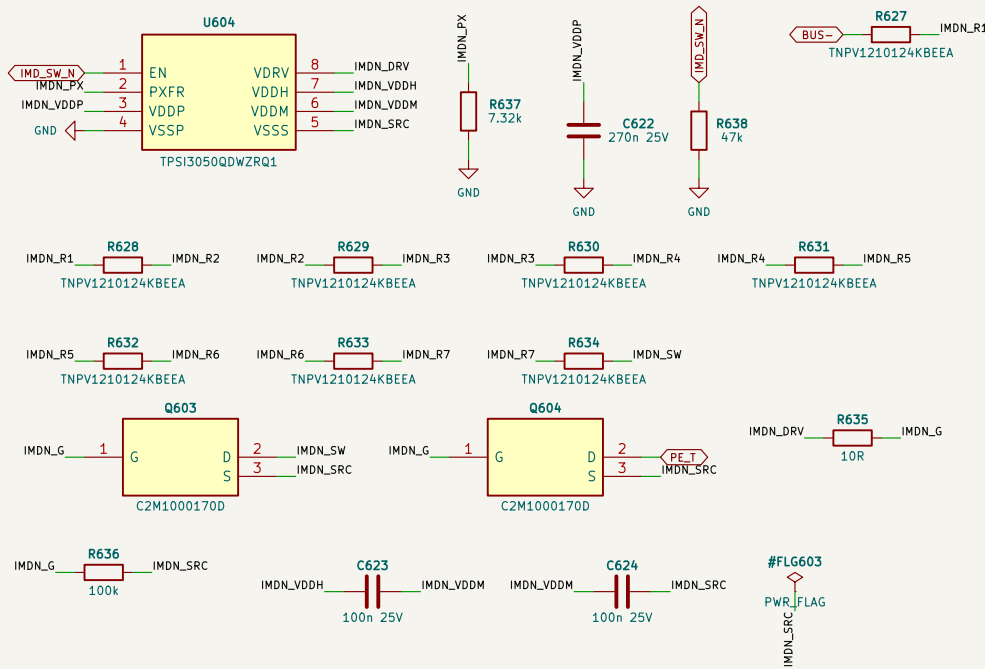
6 x 1.00 M from PE + 4.99 k: +/-1203 V linear. The divider is a fixed, known 6.005 M path to PE (included in the IMD equations, sim/port_design.py sec. 8). First element = HV–PE crossing.

TEST STRING A_X+ -> PE (IMD_SW_P)



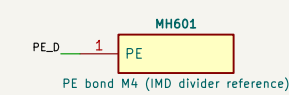
8 x TNPV1210 124 k = 992 k (125 V, 0.13 W each at 1000 V): 1 mA test current. Energised = connected. Common-source C2M1000170D pair (PA–09): blocks either polarity, so a pole below PE cannot leak through a body diode. TPSI3050 two-wire from the 24 V DO, reinforced. IDSS <= 100 uA each – EOL baseline cal.

TEST STRING BUS– -> PE (IMD_SW_N)



8 x TNPV1210 124 k = 992 k (125 V, 0.13 W each at 1000 V): 1 mA test current. Energised = connected. Common-source C2M1000170D pair (PA–09): blocks either polarity, so a pole below PE cannot leak through a body diode. TPSI3050 two-wire from the 24 V DO, reinforced. IDSS <= 100 uA each – EOL baseline cal.

PE_D BOND + ONE ACTIVE MONITOR PER DC SYSTEM



PE_D: own M4 bond (dividers only). The module disables its own measurement by leaving IMD_SW_P/N de-energised (strings open, fail-safe): a passive module still loads each pole with 6.005 M to PE.